

Sphiwe B. Skhosana

Lecturer, Department of Statistics, University of Pretoria, South Africa

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RESEARCH INTERESTS

Finite mixture models and model-based clustering; non-parametric mixtures of regressions and mixtures of experts; directional (circular) statistics; machine learning; Statistical applications in energy and environmental economics, finance, and the health sciences.

EDUCATION

2024	PhD, Mathematical Statistics , University of Pretoria. Thesis: <i>Essays on estimation strategies addressing label-switching in Gaussian mixtures of semi- and non-parametric regressions</i> (2025 Distinguished Dissertation Award, The Classification Society).
2019	MCom, Statistics , University of Pretoria (<i>with distinction</i>).
2017	BCom (Honours), Statistics , University of Pretoria.
2016	BCom, Statistics , University of Pretoria (<i>with distinction</i>).

ACADEMIC APPOINTMENTS

2020–	Lecturer (permanent, full-time), Department of Statistics, University of Pretoria.
2016, 2017	Teaching Assistant , Department of Statistics, University of Pretoria.

AWARDS AND HONOURS

2025	Distinguished Dissertation Award, The Classification Society, for the PhD thesis.
2025	Sichel Medal, South African Statistical Association: Best statistical paper published in 2024.
2025	Cluster Teaching Excellence Award in the Mathematical Sciences, Faculty of Natural and Agricultural Sciences, University of Pretoria.
2024	First prize, Published Postgraduate Paper Competition, South African Statistical Association.
2024	First prize, poster competition, Faculty of Natural and Agricultural Sciences Research Symposium, University of Pretoria.

PUBLICATIONS

Peer-reviewed journal articles

- [1] **Skhosana, S.B.**, Kanfer, F.H. and Millard, S.M. (2022). Fitting non-parametric mixture of regressions: introducing an EM-type algorithm to address the label-switching problem. *Symmetry*, 14(5), 1058.
- [2] **Skhosana, S.B.**, Millard, S.M. and Kanfer, F.H. (2023). A novel EM-type algorithm to estimate semi-parametric mixtures of partially linear models. *Mathematics*, 11(5), 1087.
- [3] **Skhosana, S.B.**, Millard, S.M. and Kanfer, F.H. (2024). A modified EM-type algorithm to estimate semi-parametric mixtures of non-parametric regressions. *Statistics and Computing*, 34(4), 125.
- [4] **Skhosana, S.B.** and Rad, N.N. (2026). Model-based clustering using a new mixture of circular regressions. *Advances in Data Analysis and Classification*, 1–30.
- [5] Harris, T., Rad, N.N. and **Skhosana, S.B.** (2026). Deep learning embedded latent class joint modelling of time-to-event and longitudinal data. *Statistical Methods in Medical Research*.

Book chapter

- [6] **Skhosana, S.B.**, Millard, S.M. and Kanfer, F.H.J. (2024). A new approach to estimate semi-parametric Gaussian mixtures of regressions with varying mixing proportions. In C.A. Coelho and D.-G. Chen (eds), *Statistical Modeling and Applications*, Emerging Topics in Statistics and Biostatistics, pp. 149–175. Springer, Cham. Co-author.

Manuscripts under review

- [7] **Skhosana, S.B.** and Yao, W. (2025). Nonparametric contaminated Gaussian mixture of regressions. *Advances in Data Analysis and Classification*.
- [8] Mambondimwewe, P., **Skhosana, S.B.** and Rad, N.N. (2025). Robust semi-parametric mixtures of linear experts using the contaminated Gaussian distribution. *Journal of Classification*.
- [9] **Skhosana, S.B.** and Nguyen, H.D. (2026). Shifted asymmetric Laplace mixtures of experts. *Econometrics and Statistics*.
- [10] **Skhosana, S.B.**, Olaniran, A.O., Rad, N.N. and Gupta, R. Economic complexity and environmental impact using a neural-network-embedded semiparametric mixture of experts model. *Energy Economics*.

Work in preparation (preliminary titles)

- [11] **Skhosana, S.B.** (est. 2026). Robust semi-parametric mixtures of regressions using scale mixtures of normal distributions, with applications in finance and economics.
- [12] **Skhosana, S.B.**, Rad, N.N., Cao, G. and Inglesi-Lotz, R. (est. 2027). Deep varying-coefficient mixtures of logistic regressions for studying energy poverty.
- [13] **Skhosana, S.B.** and Manda, S. (est. 2027). Robust and flexible mixtures for meta-analysis with applications in the health sciences.

RESEARCH FUNDING

2026	Visiting scholarship in the African Futures Research Leadership Program – travel (Return ticket to Michigan, USA) and accommodation (4 months), monthly salary and benefits (about US\$30 000).
2026	The Classification Society – travel and accommodation, invited special session, annual meeting (US\$2 000).
2024	nGAP international travel grant – COMPSTAT 2024, Giessen, Germany (R48 638).
2023	National Graduate Academy / nGAP international travel grant – CMStatistics 2023, Berlin, Germany (R45 684).
2022	University Capacity Development Programme international travel grant – CMStatistics 2022, London, UK (R54 167).
	DSI–NRF Research Development Grant, National Research Foundation, 2 years (R30 000).

SELECTED CONFERENCE PRESENTATIONS

International (contributed)

- *Estimating semi-parametric Gaussian mixtures of non-parametric regressions with an application in environmental economics.* CMStatistics 2025, Birkbeck, University of London, UK.
- *A new approach to estimate semi-parametric Gaussian mixtures of regressions with varying mixing proportions.* COMPSTAT 2024, Justus Liebig University Giessen, Germany.
- *A new approach to estimate semi-parametric Gaussian mixtures of non-parametric regressions.* CMStatistics 2023, HTW Berlin, Germany.
- *A one-step backfitting algorithm for estimating the semi-parametric mixture of partial linear models.* CMStatistics 2022, King's College London, UK.
- *A possible solution to the label-switching problem in fitting non-parametric mixture of regressions.* CMStatistics 2021, King's College London, UK.

National (contributed)

- Seminar series, Department of Economics, University of Pretoria (2024, 2025); Faculty of Natural and Agricultural Sciences Research Symposium, University of Pretoria (2024).
- SAMSA Conference, University of Pretoria (2023); Annual International Symposium on Modern Biostatistics and Machine Learning, University of Pretoria (2022, 2023).

POSTGRADUATE SUPERVISION

Current	3 PhD (co-supervisor), 10 MSc research (3 as principal supervisor), 6 Honours (1 as principal supervisor).
Completed	9 Honours (3 as principal supervisor).
PhD topics	Time series models for circular interval data; advances in latent-class joint modelling; drug discovery using artificial intelligence.
MSc topics	(as principal supervisor) Mixtures of experts; semi-parametric mixtures of non-parametric regressions; machine learning for semi-parametric mixtures of regressions with varying mixing proportions.

TEACHING

Postgraduate	MET 720 Multivariate Techniques (Honours, 2026–); STK 880 Capita Selecta (Masters, 2025).
Undergraduate	STC 137/147 Foundational Statistics (self-developed, 2025–); STK 353 Science of Data Analytics (2026–); STK 210/220 Applied Mathematical Statistics (2025); JGI 220 Statistics for Teachers (self-developed, 2020–2024); WST 153 Foundational Statistics (2023–2024); STK 110/113/120/121 (2020–2022).

COLLABORATORS, SERVICE AND MEMBERSHIP

Collaborators	N. Nakhaeirad (UP), W. Yao (UC Riverside), Y. Samadi (Southern Illinois), H.D. Nguyen (La Trobe), G. Cao (Michigan State), R. Gupta and R. Inglesi-Lotz (UP Economics), S. Manda (UP).
Service	Chair, Departmental Transformation Committee (2024); member, Faculty Ethics Committee (2025); member, Faculty Transformation Committee (2025); member, Departmental Marketing Committee (2026); Faculty Open Day (2024, 2025).
Membership	South African Statistical Association (2018–); Computational and Methodological Statistics network (2021–).

REFERENCES

- Dr S. Millard**, Head of Department, Statistics, University of Pretoria (sollie.millard@up.ac.za)
- Prof. W. Yao**, Professor of Statistics, UC Riverside (weixin.yao@ucr.edu)
- Dr H.D. Nguyen**, Assoc. Professor of Statistics, La Trobe University (h.nguyen5@latrobe.edu.au)
- Prof. R. Gupta**, Professor of Economics, University of Pretoria (rangan.gupta@up.ac.za)
- Dr N. Nakhaeirad**, Assoc. Professor of Statistics, Univ. of Pretoria (najmeh.nakhaeirad@up.ac.za)
- Prof. S. Manda**, Professor of Statistics, University of Pretoria (samuel.manda@up.ac.za)